NAME Solution

ChE 210 Quiz #2 14 February 2020

To produce ammonia (NH_3), a stream containing nitrogen (N_2) and hydrogen (H_2) in a 1:1 molar ratio is fed into a reactor. The reactor operates steadily at 100 atm and 700˚C. The single reaction taking place in the reactor is: $N_2 + 3\,H_2 \rightarrow 2\,NH_3$. The reactor converts 23% of the nitrogen entering the reactor into ammonia. There is a single output stream from the reactor. The ammonia production rate is 1200 mol/min of NH_3.

(a) Draw a labeled block diagram and calculate the degrees of freedom for the process using p. 105 of the text. Be sure to give values and a brief explanation for 4a, 4b, 5a, 5b, 5c, and 5d.

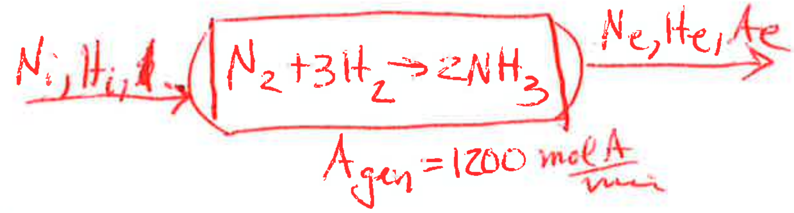
(b) Write and solve the appropriate mole balances to find the molar flow rates of all the species exiting the reactor in kmol/hr.

(c) What are the volume flow rates of the input stream to and the output stream from the reactor in m³/min?

(d) Briefly explain the physical reason for the relative magnitude of your results in part (c).

(a)

4a	5		N_i, H_i, N_e, H_e, A_e
4b	1		Haber Rx
5a	0	1	0 or A_{out}
5b	1	1	1:1 inlet ratio
5c	2	1	23% and A_{gen}
5d	3	3	3 species, 1 unit



DoF = 6 − 6

$$A_{gen} = 1200 \frac{mol\,A}{min}\ \frac{60\,min}{hr}\ \frac{kmol}{1000\,mol} = 72 \frac{kmol\,A}{hr}$$

(b) Acc = in − out + gen − cons

N_2: $0 = \dot N_i - (1-0.23)\dot N_i + 0 - \left(\frac{1\,kmol\,N_2}{2\,kmol\,NH_3}\right)\left(72\frac{kmol\,NH_3}{hr}\right) \Rightarrow \dot N_i = \frac{72}{2(0.23)}\frac{kmol}{hr} = 156.5 \frac{kmol\,N_2}{hr}$

$N_e = (1-0.23)N_i = 120.5 \frac{kmol\,N_2}{hr}$

$N_i = H_i = 156.5 \frac{kmol\,H_2}{hr}$

H_2: $0 = N_i - H_e + 0 - \frac{3\,mol\,H_2}{2\,mol\,NH_3}\ 72\frac{kmol\,NH_3}{hr} \Rightarrow H_e = N_i - \frac{3(72)}{2}\frac{kmol\,H_2}{hr} = 48.5 \frac{kmol\,H_2}{hr}$

NH_3: $0 = 0 - A_e + 72\frac{kmol\,NH_3}{hr} \Rightarrow A_e = 72 \frac{kmol\,NH_3}{hr}$

$A_i = 0$

(c) $$\dot V_i = \frac{\dot n_i RT}{P} = \frac{\left(2 \times 156.5 \frac{kmol}{hr}\right)\left(82.057 \frac{atm\,cm^3}{mol\,K}\right)(973\,K)}{(100\,atm)\left(100\frac{cm}{m}\right)^3}\ \frac{1000\,mol/kmol}{60\,min/hr}$$

$$= 4.166\ m^3/min$$

$$\dot V_e = \frac{\dot n_e}{\dot n_i}\dot V_i = \frac{(120.5 + 48.5 + 72)}{2(156.5)}\ 4.166 \frac{m^3}{min} = 3.207 \frac{m^3}{min}$$

(d) $\dot V_e < \dot V_i$ because Rx reduced total # of moles.

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Atomic Balance IN = OUT

N: $2N_i = A_e + 2N_e$

H: $2H_i = 3A_e + 2H_e$

constraint $N_i = H_i$

Performance Spec $(1-0.23)N_i = N_e$

4 eqns. 4 unks.

$$2N_i = A_e + 2(1-0.23)N_i \Rightarrow 2(0.23)N_i = A_e = 72\ \frac{kmol}{hr}$$

$$2N_i = 3A_e + 2H_e$$

$$N_i = \frac{72}{2(0.23)} = 156.5\ \frac{kmol}{hr} = H_i$$

$$H_e = N_i - \frac{3}{2}A_e$$

$$= 156.5 - \frac{3}{2}(72) = \cancel{\frac{72}{2(0.23)}}\ 48.5\ \frac{kmol}{hr}$$

$$N_e = 0.77N_i = 120.5\ \frac{kmol}{hr}$$

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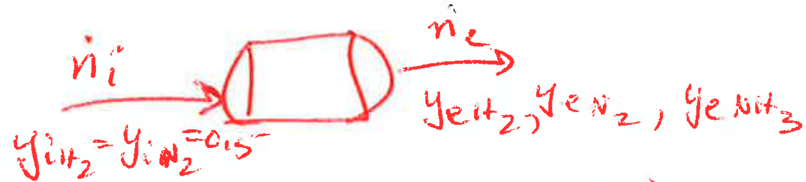

$N_2: \quad 0 = (0.5)\dot{n}_i - y_{eN_2}\dot{n}_e + 0 - (0.23)(0.5\,\dot{n}_i)$

$H_2: \quad 0 = (0.5)\dot{n}_i - y_{eH_2}\dot{n}_e + 0 - 3(0.23)(0.5\,\dot{n}_i)$

$NH_3: \quad 0 = 0 - 72\,\frac{\text{kmol } NH_3}{\text{hr}} + 2(0.23)(0.5\,\dot{n}_i)$

$y_{iN_2}\dot{n}_i = 156.5$

$y_{iH_2}\dot{n}_i = 156.5$

$\dot{n}_i = \frac{72}{0.23} = 313.0\,\frac{\text{kmol}}{\text{hr}}$

$0 = (0.77)(0.5)(313.0) - y_{eN_2}\dot{n}_e \Rightarrow y_{eN_2}\dot{n}_e = 120.5$

$0 = (0.31)(0.5)(313.0) - y_{eH_2}\dot{n}_e \Rightarrow y_{eH_2}\dot{n}_e = 48.5$

$y_{eNH_3}\dot{n}_e = 72$

$\dot{n}_e = 241$

$$\boxed{\begin{aligned} y_{eN_2} &= \frac{120.5}{241} = 0.50 \\ y_{eH_2} &= \frac{48.5}{241} = 0.20 \\ y_{eNH_3} &= \frac{72}{241} = 0.30 \end{aligned}}$$